

CounterFlag

Agentic Detection of Social-Media-Driven Market Manipulation with Adversarial Verification

Problem & why it occurs

Coordinated groups hype low-liquidity stocks on social media to inflate price, then sell into the retail-driven spike before it collapses. This persists because thin order books let small coordinated buying move price sharply; social-media and market-surveillance signals are monitored by separate systems, so the correlation that reveals manipulation is rarely made in real time; and legitimate rallies produce the same surface signature (hype + price spike), causing high false positives that make aggressive automated flagging untrustworthy.

Objective

Fuse real social and market signals to flag candidate manipulation events, then run a two-agent adversarial check — one agent builds the case, a second independently searches for an innocent explanation — before finalizing any verdict, and measure how much this reduces false positives.

System overview

Reddit/WSB posts + price/volume data → FinBERT sentiment + spike detection → XGBoost anomaly scoring → **Prosecutor agent** cites evidence for manipulation → **Defense agent** retrieves real news/filings via RAG for an innocent explanation → adjudicated verdict with both reports visible → FastAPI + Streamlit, deployed on Azure.

Dataset use & availability

All real, free: Reddit/WSB posts (Pushshift/Kaggle), price/volume (yfinance/Alpha Vantage), news & filings (NewsAPI, SEC EDGAR), and real ground truth from SEC litigation releases — supplemented with synthetic injected cases since real cases are few.

BalanceGrid

Agentic Capacity Forecasting and Solver-Grounded Pre-emptive Load Reallocation

Problem & why it occurs

Telecom operators manage congestion reactively — capacity is adjusted only after service already degrades. This happens because traffic spikes in one cell often have unused slack in a neighboring cell that per-cell alerting doesn't capture; forecasting and reallocation are handled as separate processes

bridged by a slow manual step; and reallocation is a real constrained-optimization problem that simple thresholds get wrong in both directions.

Objective

Forecast short-horizon per-cell congestion, diagnose whether a spike is routine or anomalous, and — for high-confidence routine cases — have a constrained solver compute a feasible reallocation, closing the loop from forecast to action.

System overview

Milan CDR data → per-cell forecasting (LightGBM/PatchTST) → historical-percentile thresholding → **Diagnosis agent** classifies routine vs. anomalous using neighboring-cell correlation + the real event calendar → **OR-Tools solver** computes the reallocation (never the LLM) → anomalies escalate to a human → FastAPI + Streamlit/Plotly map, deployed on Azure.

Dataset use & availability

Real, free: Telecom Italia Milano Grid CDR dataset (Kaggle) — genuinely real aggregated cell-level activity — plus the real Milan public event/holiday calendar for diagnosis validation. Scoped to traffic-volume congestion, not deep packet/RAN telemetry.